

Corner-Adapted Quadrature for Rellich Normalization

Why the current rule fails at reentrant corners, and what replaces it

lappy development notes

Abstract

The Rellich identity computes $\|u\|_{L^2(\Omega)}$ from boundary Cauchy data alone. Its integrand carries a factor $r^{2\nu-2}$ at a corner of interior angle α , where $\nu = \pi/\alpha$; at a reentrant corner $\nu < 1$ and this is singular. We show (i) why the singularity can be removed *geometrically* by placing the Rellich reference point x_0 at the corner, (ii) why the current graded rule cannot integrate it when x_0 is placed elsewhere — the failure is truncation, not under-resolution — and (iii) that a corner-local Gauss–Jacobi rule in the variable $t = r^\nu$ integrates it to machine precision with $O(10)$ nodes, removing the constraint on x_0 altogether. A caveat is identified: the substitution rationalizes the corner expansion completely only when $2/\nu \in \mathbb{Z}$, which holds for the 270° corner but not generically.

1 The identity and its corner integrand

For $-\Delta u = \lambda u$ in $\Omega \subset \mathbb{R}^2$ with $u = 0$ on $\partial\Omega$, the Rellich–Pohozaev identity with reference point $x_0 \in \mathbb{C}$ gives the L^2 norm as a pure boundary integral,

$$\|u\|_{L^2(\Omega)}^2 = \frac{1}{2\lambda} \int_{\partial\Omega} ((x - x_0) \cdot n) \left(\frac{\partial u}{\partial n} \right)^2 ds = \frac{1}{2\lambda} \int_{\partial\Omega} r_N \left(\frac{\partial u}{\partial n} \right)^2 ds, \quad (1)$$

with $r_N := (x - x_0) \cdot n$. The identity is exact for *every* x_0 ; the choice of x_0 affects only how hard the integral is to evaluate.

1.1 Local structure at a corner

Let the corner sit at the origin with interior angle α and edges along $\theta = 0$ and $\theta = \alpha$, and set

$$\nu := \frac{\pi}{\alpha}, \quad \nu_k := k\nu = \frac{k\pi}{\alpha}, \quad k = 1, 2, \dots \quad (2)$$

The corner is *reentrant* when $\alpha > \pi$, i.e. $\nu < 1$; the 270° corner has $\nu = 2/3$ and the slit limit $\alpha \rightarrow 2\pi$ gives $\nu \rightarrow 1/2$. Any Dirichlet solution admits the local expansion

$$u(r, \theta) = \sum_{k \geq 1} c_k J_{\nu_k}(\sqrt{\lambda} r) \sin(\nu_k \theta). \quad (3)$$

On the edge $\theta = 0$ the outward normal is $n = -e_\theta$, so $u \equiv 0$ there forces the normal derivative to come entirely from the angular derivative:

$$\left. \frac{\partial u}{\partial n} \right|_{\theta=0} = -\frac{1}{r} \left. \frac{\partial u}{\partial \theta} \right|_{\theta=0} = -\sum_{k \geq 1} c_k \nu_k \frac{J_{\nu_k}(\sqrt{\lambda} r)}{r}. \quad (4)$$

Using $J_\mu(z) = (z/2)^\mu A_\mu(z^2)$ with A_μ entire and $A_\mu(0) \neq 0$,

$$\left. \frac{\partial u}{\partial n} \right|_{\theta=0} = r^{\nu-1} F(r), \quad F(r) = \sum_{k \geq 1} b_k r^{(k-1)\nu} A_{\nu_k}(\lambda r^2), \quad (5)$$

so that

$$\left(\frac{\partial u}{\partial n} \right)^2 \Big|_{\theta=0} = r^{2\nu-2} G(r), \quad G(r) = F(r)^2 = \sum_{p \geq 0} \sum_{q \geq 0} g_{pq} r^{p\nu+2q}. \quad (6)$$

Two families of exponents appear in G : the *corner* exponents $r^{p\nu}$ generated by the expansion (3), and the *Bessel* exponents r^{2q} from the entire factors. This double structure is the crux of Section 3.1.

1.2 The geometric escape: $r_N \equiv 0$

Proposition 1. *If x_0 is placed at the corner, then $r_N \equiv 0$ on both straight edges emanating from it.*

Proof. On the edge $\theta = 0$, $x - x_0 = re^{i \cdot 0}$ is parallel to the edge, while $n = -e_\theta$ is perpendicular to it; hence $(x - x_0) \cdot n = 0$ pointwise, and likewise on $\theta = \alpha$. $\square$

The singular factor $r^{2\nu-2}$ is therefore multiplied by an *exact* zero, and the corner contributes nothing: the boundary integral collapses onto the parts of $\partial\Omega$ where u is smooth. This is why placing x_0 at a reentrant corner yields machine precision with any reasonable quadrature. It is also why the escape does not generalize: x_0 is a single point, and a domain with several reentrant corners (H-shape, GWW) cannot zero r_N at all of them.

2 Why the graded rule fails for x_0 off the corner

For general x_0 , r_N is a nonzero constant along each straight edge (it is the signed distance from x_0 to the edge's line), and the edge contributes

$$I_{\text{edge}} = \rho \int_0^R r^{2\nu-2} G(r) dr, \quad \rho := r_N = \text{const.} \quad (7)$$

This converges for $\nu > 1/2$, but the convergence is deceptive.

Lemma 1 (Mass concentration). *The fraction of $\int_0^R r^{2\nu-2} dr$ contributed by $r < \varepsilon$ is $(\varepsilon/R)^{2\nu-1}$.*

The exponent $2\nu - 1$ vanishes as $\nu \rightarrow 1/2$. Numerically, with $R = 1$ and $\varepsilon = 10^{-9}$:

α	ν	$2\nu - 1$	mass below 10^{-9}
1.25π	0.800	0.600	4.0×10^{-6}
1.5π	0.667	0.333	1.0×10^{-3}
1.75π	0.571	0.143	5.2×10^{-2}
1.9π	0.526	0.053	0.336
1.99π	0.503	0.005	0.901

For a near-slit corner, *ninety percent of the integral lives within 10^{-9} of the corner*. lappy’s Kress-graded rule clamps its innermost node at `quad._KRESS_TAU_FLOOR` = 10^{-9} (a guard against a genuine defect: nodes built as $p_0 + \tau L \hat{d}$ in absolute coordinates round to exactly p_0 once τL falls below machine epsilon relative to $|p_0|$). The rule is thus *truncated*, not under-resolved: refining from 565 to 6790 nodes leaves the innermost node at 10^{-9} and the error unmoved. Observed errors track Lemma 1 over eight orders of magnitude, scaling linearly in $|x_0 - \text{apex}|$ with constant $\approx (10^{-9})^{2\nu-1}$.

Remark 1. The floor is a symptom of *coordinate representation*, not of floating-point resolution. The value $r = 10^{-30}$ is perfectly representable; it is $p_0 + \tau L \hat{d}$ that is not. A rule evaluated in corner-local (r, θ) never forms that difference. This distinction is what makes Section 3 possible.

3 A corner-local Gauss–Jacobi rule

Gauss–Jacobi quadrature with weight $(1-x)^a(1+x)^b$ integrates $\int_{-1}^1 (1-x)^a(1+x)^b f(x) dx$ exactly for f polynomial of degree $\leq 2n-1$, and spectrally for f analytic. Mapping $r = R(1+x)/2$ and matching $b = \beta := 2\nu - 2$ to (7),

$$\int_0^R r^{2\nu-2} G(r) dr = \left(\frac{R}{2}\right)^{2\nu-1} \sum_{i=1}^n w_i G(r_i), \quad r_i = \frac{R}{2}(1+x_i). \quad (8)$$

Admissibility requires $\beta > -1$, i.e. $\nu > 1/2$: precisely the range of integrability, degenerating exactly at the slit.

Two properties matter as much as the accuracy:

1. **The nodes stay away from the corner.** The smallest Jacobi node sits at distance $O(n^{-2})$, not 10^{-9} : for $n = 16$ the innermost node is near 10^{-3} . The singularity is absorbed *analytically* into the weight rather than resolved by clustering, so neither the truncation floor nor the coordinate cancellation of Remark 1 ever arises.
2. **Only ν is needed.** $G(r) = (\partial_n u)^2 r^{2-2\nu}$ is formed numerically from black-box Cauchy data; no analytic knowledge of u is used. Since $\nu = \pi/\alpha$ comes exactly from the geometry, this is free — but it must be exact, see Section 4.

3.1 The multi-exponent obstruction and the substitution

Rule (8) is spectral only if G is analytic. By (6) it is not: G carries fractional powers $r^{p\nu}$. For a *single* mode ($c_k = 0$, $k \geq 2$) we have $G(r) = b_1^2 A_{\nu_1}(\lambda r^2)^2$, analytic in r^2 , and (8) is spectral — but a single mode is not the situation of interest. For a general solution, convergence degrades to algebraic.

Proposition 2 (Rationalizing substitution). *Under $t = r^\nu$, equivalently $r = t^{1/\nu}$, the edge integral becomes*

$$\int_0^R r^{2\nu-2} G(r) dr = \frac{1}{\nu} \int_0^{R^\nu} t^{1-1/\nu} \tilde{G}(t) dt, \quad \tilde{G}(t) = \sum_{p,q} g_{pq} t^{p+2q/\nu}, \quad (9)$$

a Jacobi-weight integral with exponent $\beta' = 1 - 1/\nu$. The corner family $r^{p\nu}$ maps to the integer powers t^p .

Proof. $dr = \frac{1}{\nu} t^{1/\nu-1} dt$ and $r^{2\nu-2} = t^{2-2/\nu}$; the exponents add to $(2-2/\nu) + (1/\nu-1) = 1-1/\nu$. Substituting $r^{p\nu+2q} = t^{p+2q/\nu}$ into (6) gives $\tilde{G}$. $\square$

Admissibility $\beta' > -1$ again requires exactly $\nu > 1/2$.

The caveat. The substitution rationalizes the corner family completely, but maps the Bessel family $r^{2q} \mapsto t^{2q/\nu}$, which is integral only when $2/\nu \in \mathbb{Z}$ — equivalently $\alpha = 2\pi/m$ for an integer m . The 270° corner is such a case:

$$\alpha = \frac{3}{2}\pi \implies \nu = \frac{2}{3} \implies 2/\nu = 3 \in \mathbb{Z}, \quad \tilde{G}(t) = \sum g_{pq} t^{p+3q} \text{ (a power series in } t\text{)}. \quad (10)$$

For generic α the residual exponents $2q/\nu$ are fractional, but they are *high order*: the leading one is $2/\nu > 2$, so $\tilde{G} \in C^2$ at worst and Gauss convergence is algebraic of high order rather than spectral. This is confirmed in Section 5(c): the machine precision at $n = 8$ is specific to $\alpha = \frac{3}{2}\pi$, while generic α converges at roughly two digits per node doubling — still reaching 10^{-12} by $n = 32$, which is cheap enough that the distinction is of no practical consequence.

4 Sensitivity to ν

The weight exponent must use ν at full precision. Perturbing $\nu \rightarrow \nu + \delta$ leaves a residual factor $r^{-2\delta}$ in the integrand, which is unbounded as $r \rightarrow 0$ for $\delta > 0$ and destroys the analyticity the rule depends on. Measured at $\alpha = 1.5\pi$ (true $\nu = 2/3$), mode $(1, 1)$:

ν used in β	28 nodes	192 nodes
0.6667 (exact)	-7.5×10^{-15}	-3.1×10^{-15}
0.667	6.6×10^{-5}	1.7×10^{-5}
0.600	-1.8×10^{-2}	-4.7×10^{-3}

A relative error of 3×10^{-4} in ν costs four digits and reduces convergence to algebraic. Since $\nu = \pi/\alpha$ is exact from the geometry this costs nothing — but it rules out tabulated, rounded, or margin-padded exponents of the kind used for grading orders.

5 Numerical verification

All errors are relative, against closed-form sector eigenfunctions of unit norm (exact L^2 norms, so the reference is analytic rather than quadrature-based).

(a) Single mode, 270° corner, x_0 at the failing default. Cauchy data treated as a black box; $2n_{\text{edge}} + n_{\text{arc}}$ nodes total.

nodes	Gauss–Jacobi	existing Kress rule
16	-7.1×10^{-9}	—
28	-7.5×10^{-15}	—
2264	—	2.2×10^{-7} (plateau)

(b) Multi-exponent corner data. A five-term expansion (3) times an analytic tail, integrated against an mpmath reference at 40 digits, at $\nu = 2/3$:

n	Jacobi in r	Jacobi in $t = r^\nu$
8	-4.6×10^{-3}	4.0×10^{-15}
16	-1.2×10^{-3}	7.3×10^{-15}
64	-7.2×10^{-5}	1.6×10^{-13}

Plain Jacobi in r is algebraic, as Section 3.1 predicts; the substituted rule is spectral and *degrades* slowly past $n \approx 32$ from roundoff amplification, so small n is both cheaper and more accurate.

(c) Generic α : is $\nu = 2/3$ special? The same five-term data, substituted rule, against an *exact* reference (see the warning below).

α	$2/\nu$	$n = 8$	$n = 16$	$n = 32$	$n = 64$
1.5π	3.000 (int.)	2.0×10^{-15}	2.0×10^{-15}	-2.6×10^{-14}	1.7×10^{-13}
1.33π	2.667	2.8×10^{-8}	2.7×10^{-10}	2.7×10^{-12}	5.1×10^{-14}
1.25π	2.500	4.7×10^{-8}	5.1×10^{-10}	5.9×10^{-12}	5.0×10^{-14}
1.6π	3.200	-5.6×10^{-9}	-3.4×10^{-11}	-1.6×10^{-13}	-7.7×10^{-13}
1.7π	3.400	-5.2×10^{-9}	-2.7×10^{-11}	-3.4×10^{-13}	-9.8×10^{-13}
1.85π	3.700	-1.5×10^{-9}	-6.0×10^{-12}	1.9×10^{-14}	-2.5×10^{-12}
1.95π	3.900	-1.4×10^{-10}	-2.0×10^{-13}	-4.0×10^{-13}	-1.2×10^{-11}

Only $\alpha = \frac{3}{2}\pi$ is machine-exact at $n = 8$, confirming Section 3.1: the integer $2/\nu$ is genuinely special. Every other angle converges at $\approx$ two digits per doubling — high-order algebraic, exactly as predicted — and all reach 10^{-12} or better by $n = 32$. The mild degradation past $n \approx 32$ is roundoff, not truncation.

Warning: mpmath’s quad is not a safe reference here. The endpoint behaviour $r^{2\nu-2} \rightarrow r^{-1}$ as $\nu \rightarrow \frac{1}{2}$ defeats `mp.quad` even at 40 digits. Measured against the exact termwise value $\int_0^1 r^a dr = (a+1)^{-1}$, its error reaches -4.7×10^{-5} at $\alpha = 1.85\pi$ and -4.3×10^{-2} at $\alpha = 1.95\pi$. Using it as ground truth produces convincing but entirely spurious “plateaus” in the rule under test. Any reference for this integral should be closed-form.

A trap worth recording. At $\alpha = 3\pi/2$ the two edges carry $r_N = \text{Im } x_0$ and $-\text{Re } x_0$ against identical $(\partial_n u)^2$, so any x_0 on the diagonal $\text{Re } x_0 = \text{Im } x_0$ cancels the singular contribution *identically*. A test using such an x_0 silently measures the smooth part of the boundary only. Corner-quadrature experiments should always report the edge/arc split.

6 Recommended construction

For each corner with $\nu < 1$, and each straight edge meeting it:

1. Work in corner-local coordinates (r, θ) ; never form $p_0 + \tau L \hat{d}$.

2. Panel the edge, substitute $t = r^\nu$ with $\nu = \pi/\alpha$ exact.
3. Apply Gauss–Jacobi with $\beta' = 1 - 1/\nu$ and $n = 8\text{--}16$ nodes.

Smooth portions of $\partial\Omega$ retain plain Gauss–Legendre. Edges meeting two singular corners are split, one panel per endpoint.

This removes the requirement that x_0 sit at a corner, and with it the obstruction to multi-corner domains: no partition of unity over corners is needed. The requirement $\nu > 1/2$ is not restrictive — it is exactly the condition for (7) to converge at all — but the slit limit $\nu \rightarrow 1/2$ remains delicate, both weights degenerating as $\beta, \beta' \rightarrow -1$.

Not yet verified. A real `FourierBesselBasis` through this rule (as opposed to closed-form or synthetic data), and a genuinely multi-corner domain.